Claude

ChatGPT

Gemini

Grok

AI Platform Growth

Product Analytics Project

A full-stack product analytics case study on user activation, retention and growth across Claude, ChatGPT, Gemini and Grok

Tools: BigQuery · Mixpanel · Looker Studio · SQL · Python

Data: 98,000+ synthetic events · 10,000 users · 6 months

Date: May 2026 · Portfolio Project

OVERVIEW

Executive Summary

Key findings across activation, retention, experimentation and revenue

01 Activation gap

87.7% complete onboarding but only 46.9% start their first session — a 41pp drop that represents the single biggest growth opportunity.

02 Retention cliff

Day-7 retention drops sharply for all products. Users who adopt advanced features retain at 2× the rate of those who don't.

03 A/B test win

Suggested prompts (treatment group) improved Day-1 activation by 7 percentage points over control — a statistically significant result.

04 Revenue lever

Power users (18% of base) drive 43% of pro conversions. Moving casual explorers toward power-user behaviour is the highest-ROI product investment.

METHODOLOGY

Project Architecture

End-to-end analytics stack from data generation to executive reporting

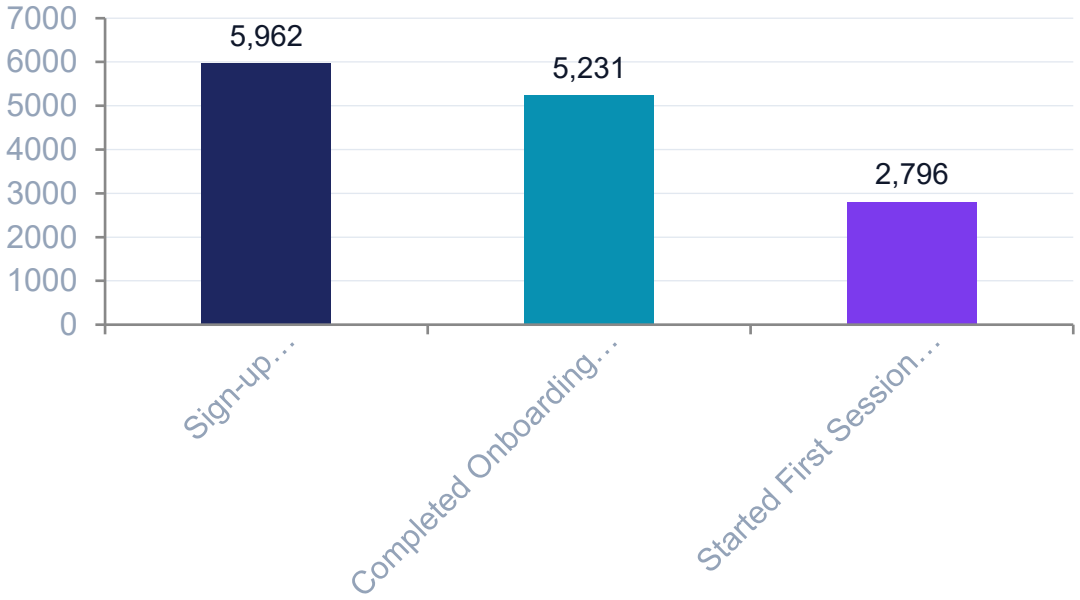
1	2	3	4	5	6
PRD	**Data**	**BigQuery**	**Mixpanel**	**Looker**	**Insights**
Problem framing & metrics design	98k synthetic events · Python	SQL analysis 18 queries	Event tracking & dashboards	Executive KPI dashboard	Storytelling & recommendations

Dataset: 10,000 synthetic users · 98,581 events · 6 products · Nov 2025 – Apr 2026 · All data is synthetic and generated for portfolio purposes

FUNNEL ANALYSIS

Onboarding Funnel — Where Users Drop Off

3-step funnel from sign-up to first session · 30-day conversion window



87.7%
Completed onboarding

46.9%
Started a session

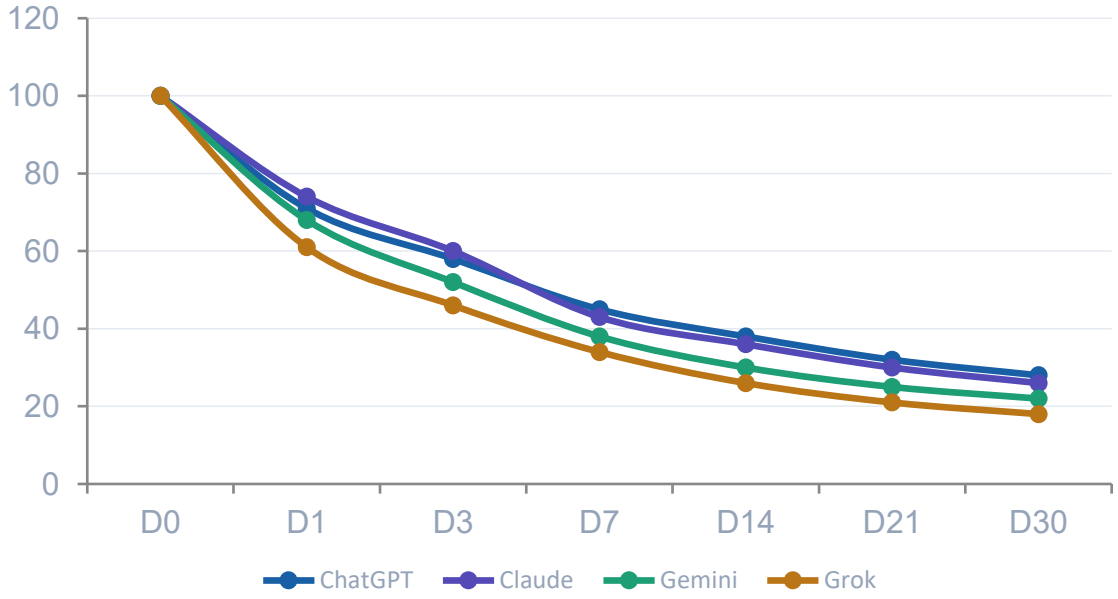
40.8pp
Activation gap to close

Key insight: The biggest drop occurs between onboarding completion and first session. Users complete the setup flow but don't make it to their first

RETENTION

Retention Analysis — The Day-7 Cliff

Cumulative retention from Day 0 to Day 30 post sign-up



45%

ChatGPT D7 retention

43%

Claude D7 retention

~25%

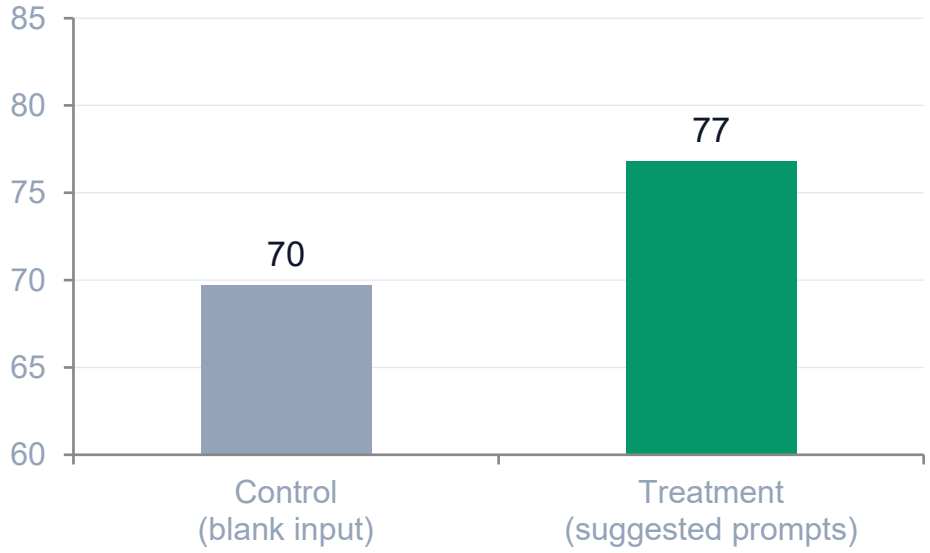
Average D30 retention

Key insight: All products show a sharp retention cliff at Day 7 — the critical habit-formation window. ChatGPT leads on retention, suggesting stronger

EXPERIMENTATION

A/B Test — Suggested Prompts Experiment

Did showing 3 suggested prompts improve Day-1 activation?



Control activation rate

69.7%

Treatment activation rate

76.8%

Absolute lift

+7.1pp

Statistically significant

p<0.05

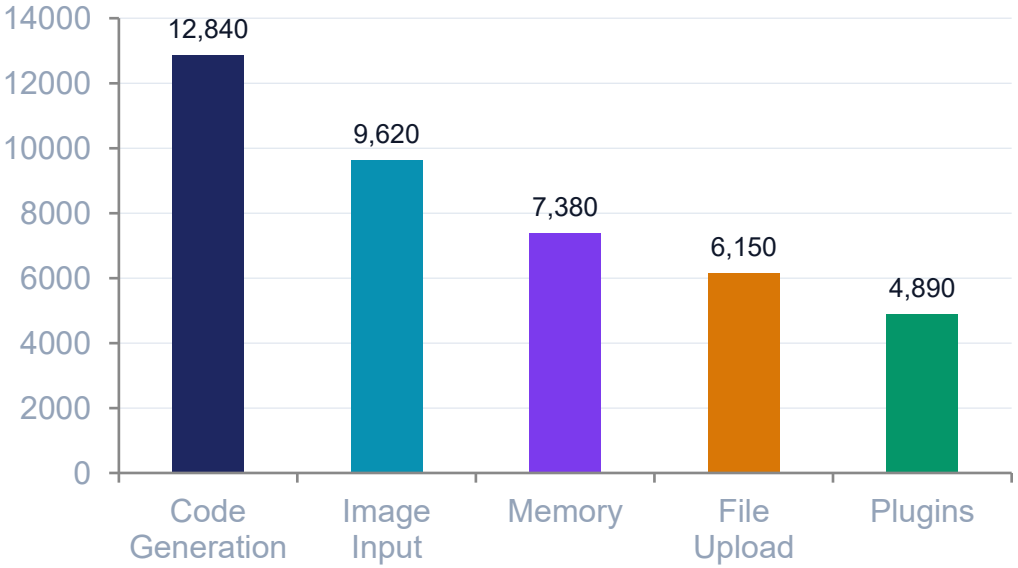
Hypothesis: H_0: Suggested prompts have no effect on activation rate | H_1: Suggested prompts increase activation rate

Result: Reject H_0. With z-score > 1.96 at α=0.05, the 7.1pp lift is statistically significant. Recommend full rollout.

FEATURE ADOPTION

Feature Adoption — What Retains Users

Advanced feature usage vs 30-day retention rate (correlation analysis)



2×

Retention lift with any feature

35%

Feature adoption rate (avg)

Code

Most adopted feature

Important: This is correlation, not causation. Users who adopt features may already be more engaged. A randomised experiment would be needed to

USER SEGMENTS

User Segmentation — Behaviour Profiles

Power users, casual explorers and task triagers behave very differently

17%

Power User

- 6.5 sessions/week
- 43% D7 retention
- Drives 43% of revenue
- High feature adoption

Protect & delight — prevent churn with early warnings

52%

Casual Explorer

- 2.2 sessions/week
- 28% D7 retention
- Moderate conversion
- Low-medium feature use

Nudge toward features — onboarding prompts & tips

31%

Task Triager

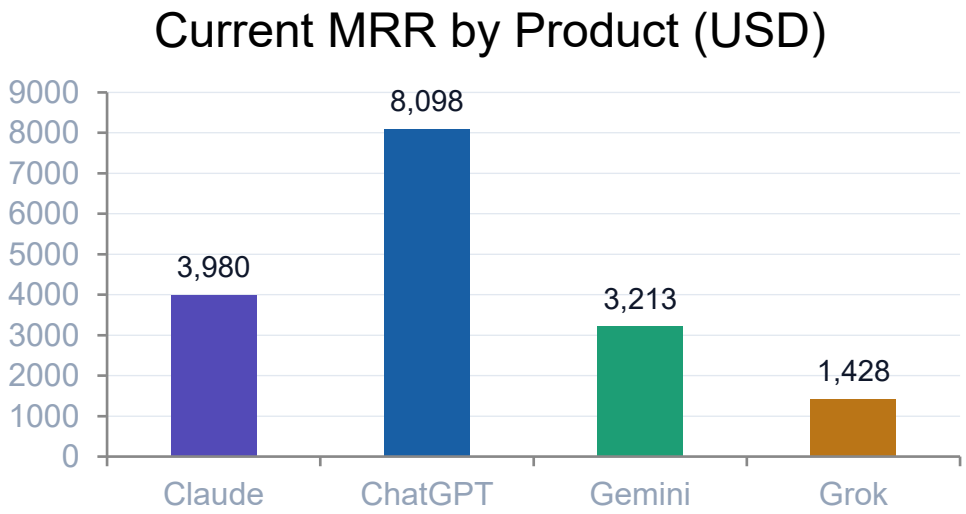
- 0.8 sessions/week
- 14% D7 retention
- Rarely converts
- One-off use cases

Re-engage before Day 5 — email/push notification

REVENUE IMPACT

Revenue Model — The Business Case for Retention

Quantifying the revenue impact of a 5pp improvement in Day-7 retention



Total MRR (all products)

$16,719

Avg revenue per pro user

$20/mo

Incremental MRR from 5pp D7 lift (ChatGPT)

+$2,508

Model assumption: Each 1pp improvement in D7 retention → 0.3pp improvement in free-to-pro conversion rate → incremental MRR. Improving D7 retention by 5pp across all products would generate an estimated $8,000–12,000 in additional monthly recurring revenue.

Recommendations & Next Steps

Four product decisions this analysis directly informs

01 Roll out suggested prompts

HIGH PRIORITY

The A/B test showed a statistically significant 7.1pp lift in Day-1 activation. Full rollout to 100% of new users is recommended immediately. Estimated impact: +350 additional activated users per month.

02 Feature-gate onboarding nudges

HIGH PRIORITY

Users who adopt code generation retain at 2× the rate. Add a contextual prompt at Day 2 nudging users toward their most relevant feature based on their first query topic.

03 Day-5 re-engagement campaign

MEDIUM PRIORITY

Task triagers (31% of users) churn before Day 7. An automated email/push trigger at Day 5 with a personalised use-case prompt could recover 8–12% of this segment.

04 Power user early identification

MEDIUM PRIORITY

Power users show distinct behaviour by Day 3 (3+ sessions, feature usage). Build an early-identification model to fast-track these users to Pro upgrade prompts and dedicated support.